Correction to "3 × 3 idempotent matrices over some domains and a conjecture on nil-clean matrices".

July 27, 2026

In Theorem 2.2, the ID, GCD hypothesis should be removed and replaced by Bézout (commutative domain).

Comment. On p.94 we recall the following: "Simplifying the writing, we can suppose E has one of the three following forms: $[C, sC, vC]$, $[0, C, sC]$, $[0, 0, C]$ where s and v are elements of R and C is a column with at least one nonzero entry".

Without this simplification, $E = [sC, tC, vC]$ with a column $C = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Indeed, *over commutative Bézout domains*, the row rank, the column rank and the inner rank of any matrix are all equal (see Theorem 4.4, [1]). Hence $E = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \begin{bmatrix} s & t & v \end{bmatrix}$ and $Tr(E) = 1$ iff $as + bt + cv = 1$.

As for the simplified cases above, it can be checked that E is idempotent.

The Bézout hypothesis on commutative domains can be further generalized to the so-called *Sylvester domains* (see [1], **5.5**).

References

[1] P. M. Cohn *Free Rings and Their Relations*. 2nd ed., Academic Press, 1985.